SMART TRAFFIC MANAGEMENT SYSTEM

Testing Document: Test Plan, Test Cases, Results and Defect Log

Item	Value
System under test	STMS multi-agent simulator (src/), demo runner, scenarios S1-S8
Test suites	tests/test_unit.py (23) | tests/test_integration.py (9) | tests/test_scenarios.py (9)
Result	43 / 43 PASSED, 0 failed, 0 errors (18 s, Python 3.11.15)
Evidence	results/test_report.json (generated by python tests/run_tests.py)
Prepared by	Team STMS (M1-M5); test lead: Member 5

1. Purpose and Scope

This document shows that the system does what the report claims. Every claim in the report maps to at least one automated test with an explicit pass criterion. The document covers:

- correctness of each algorithm in isolation (unit)
- correctness and **safety** of agent-to-agent interaction inside a running simulation (integration)
- whether each evaluation scenario meets its acceptance criterion (system)

Out of scope: real-world calibration, communication failures and turning movements (see the report's limitations).

1.1 How to run

```
pip install -r requirements.txt
python tests/run_tests.py                          # all 43 tests + JSON report
python -m unittest tests.test_unit -v              # one suite
python -m unittest
tests.test_integration.TestAgentInteraction.test_IT02_yellow_safety_invariant
```

2. Test Strategy

Level	What it checks	Technique	Count
Unit (UT)	A*, BFS, UCS, Greedy; heuristic properties; BPR, capacity, headway; Max-Pressure decisions; broker; hill climbing; vehicle beliefs	Known-answer tests, property checks over every node/edge, worked examples	23
Integration (IT)	Message protocols between agents, safety and conservation invariants, determinism	Run the simulator and assert on the message history and every tick's state	9
System (ST)	Each scenario S1-S8 meets its acceptance criterion	Seeded Poisson demand, baseline vs treatment on identical demand, averaged over seeds, run until every vehicle arrives	9

Principles.

- **Fair baselines:** the baseline and the treatment always get identical demand (same seed), and only one thing changes between them (for example, preemption on or off with the same controller).
- **No survivorship bias:** scenarios run until the last vehicle arrives; vehicles that never arrive are counted, not dropped (for example, 'starved' cars in S2).
- **Invariants over examples:** safety is checked on every phase change of a 400 s run, not on a single hand-picked moment.
- **Reproducibility:** fixed seeds; IT-09 proves two runs are identical.

2.1 Traceability: objectives to tests

Objective / claim	Rubric item	Tests
Adaptive decentralised signal control beats fixed-time	R1 Algorithmic modelling; R2 Demo & testing	UT-13..18, ST-01, ST-09
No approach is starved	R1 Algorithmic modelling	UT-16, ST-03
A* is optimal and efficient (admissible, consistent)	R1 Search strategy	UT-01..08, ST-07

Objective / claim	Rubric item	Tests
Emergency corridor is fast and safe	R2 Multi-agent interaction	IT-01, IT-02, ST-04
Bus priority is negotiated, not imposed	R2 Multi-agent interaction	IT-03, IT-04, ST-06
Incident information changes vehicle plans	R2 Multi-agent interaction	UT-23, IT-05, ST-05
Coordinator improves controllers (hill climbing)	R1 Algorithmic modelling	UT-21, IT-06, ST-08
Peer cooperation (green-wave)	R2 Multi-agent interaction	IT-07
Physical model is sound (capacity, headway, conservation)	R2 Code structure	UT-09..12, IT-08
Scales with grid size	R2 Scalability	UT-12, ST-09
Results are reproducible	R2 Demo & testing	IT-09

3. Test Cases and Results

Expected is the pass criterion encoded in the test; **Actual** is the value the test measured on the final run. System tests use fewer seeds than the report (to keep the suite under a minute), so their actual values differ slightly from the report's tables while meeting the same criteria.

3.1 Unit tests

ID	Owner	Expected (pass criterion)	Actual	Result
UT-01	M3	h(n) <= true optimal cost h*(n) for every node and goal tested	max h/h* ratio = 1.000 (<= 1 required)	PASS
UT-02	M3	h(u) <= c(u,v) + h(v) for every road and goal (consistency / monotonicity)	501 (edge, goal) pairs satisfy the triangle inequality	PASS
UT-03	M3	A* returns the same optimal cost as UCS/Dijkstra on 60 random trips	60 / 60 trips: cost(A*) == cost(UCS)	PASS
UT-04	M3	A* never expands more nodes than UCS (same tie-breaking)	total expansions A* = 924, UCS = 2377	PASS
UT-05	M3	when A->B is closed, A* routes A->D->C	path before: A-B-C, after closure: A-D-C	PASS
UT-06	M3	BFS returns a fewest-roads path	W0->E0 on 6x6 grid: 7 roads (minimum = 7)	PASS
UT-07	M3	search reports failure (empty path, infinite cost) when the goal is cut off	all 4 algorithms return [] for an unreachable goal	PASS
UT-08	M3	Greedy best-first returns a valid (connected) path, possibly sub-optimal	20 / 20 greedy paths use only existing roads	PASS
UT-09	M1	BPR travel time grows with load, equals free-flow when empty, infinite when closed	empty 20.0s, full 23.0s, closed inf	PASS
UT-10	M1	a road refuses vehicles when full or closed	3rd vehicle refused at capacity 2; refused when closed	PASS
UT-11	M1	stop line discharges at most one vehicle every 2 s (1800 veh/h/lane)	discharge blocked at +1 s, allowed at +2 s	PASS
UT-12	M1	an R x C grid has R*C intersection agents and the right number of roads	2x2, 3x4, 6x6 grids: agent and road counts correct	PASS
UT-13	M2	after min-green, switch (via YELLOW) when the red approach has higher pressure	EW green, NS pressure 20+10 vs EW 2 -> YELLOW	PASS
UT-14	M2	never switch before the minimum green time	at 2 s (< 5 s min green) phase held	PASS
UT-15	M2	forced switch at max-green if the other approach has waiting cars	NS green 30 s with 1 EW car waiting -> YELLOW	PASS
UT-16	M2	no switch towards an approach with no waiting vehicles (no wasted green)	EW empty -> NS kept green even past max-green	PASS
UT-17	M2	yellow lasts yellow_time and is followed by the OPPOSITE green	yellow held at 2 s, NS->YELLOW->EW at 3 s	PASS
UT-18	M2	P = sum max(0, Q_in - Q_out) over movements (worked example)	movements (8,2),(3,5) -> 6; EW with 10 s red -> 2 + 5 = 7	PASS
UT-19	M5	point-to-point delivery and reply threading (same conversation id)	REQUEST/AGREE share conversation C010323	PASS
UT-20	M5	broadcast reaches everyone except the sender; unknown receiver is reported	B and C received, A (sender) did not; unknown receiver -> False	PASS
UT-21	M5	hill climbing reaches the global minimum of a convex test function	found (7,2) from (0,9) in 15 iterations	PASS
UT-22	M3	a vehicle is not on the road before its departure time	not departed at t=19, departed at t=21	PASS
UT-23	M3	learning ROAD_BLOCKED makes the road cost infinite for that vehicle; ROAD_CLEARED restores it	cost inf after ROAD_BLOCKED, finite after ROAD_CLEARED	PASS

3.2 Integration tests

ID	Owner	Expected (pass criterion)	Actual	Result
IT-01	M4	ambulance PREEMPT_OVERRIDE -> intersection AGREE -> ambulance CANCEL after crossing, per intersection	INT_10 and INT_00 each: PREEMPT_OVERRIDE, AGREE, CANCEL; ambulance wait 0 s	PASS
IT-02	M4	SAFETY - no signal ever goes directly from one green to the other (always via YELLOW), even with preemption and bus priority	202 phase changes checked: 0 green->green, 0 yellows shorter than 3 s	PASS
IT-03	M4	bus priority is ACCEPTED on a light cross street and REJECTED on a heavy one	cross queue 0 -> ACCEPT_PROPOSAL; cross queue 10 -> REJECT_PROPOSAL	PASS
IT-04	M4	an ambulance preemption outranks bus priority (bus proposal rejected)	bus got REJECT_PROPOSAL: 'emergency preemption active'	PASS
IT-05	M5	TMC INCIDENT_ALERT makes every affected vehicle drop the closed road from its plan	6 / 6 vehicles re-planned away from INT_00->INT_01 and arrived	PASS
IT-06	M5	coordinator INFORM PARAMETER_UPDATE changes every intersection's controller	all 4 intersections now min_green 8 s, max_green 40 s	PASS
IT-07	M5	releasing a platoon sends INFORM PLATOON_INCOMING to the downstream neighbour, which stores it	1 PLATOON_INCOMING messages exchanged on a 1x3 corridor	PASS
IT-08	M5	every vehicle is on at most one road, no road exceeds capacity, and all vehicles arrive	180 vehicles, all arrived at t=397 s; peak road occupancy 35%	PASS
IT-09	M5	the same seed gives exactly the same result (reproducible experiments)	two runs identical: avg wait 8.34 s, 21 messages	PASS

3.3 System / acceptance tests

ID	Owner	Expected (pass criterion)	Actual	Result
ST-01	M2	Max-Pressure reduces average wait by at least 25% vs Fixed-Time, all vehicles arrive	15.36 s -> 8.2 s (46.6% less)	PASS
ST-03	M2	with anti-starvation no minority vehicle is left unserved; without it some are	starved E-W cars: with = 0, without = 10	PASS
ST-04	M4	preemption gives the ambulance zero signal delay, >= 20% faster, < 1 s cost to others	54.0 s -> 38.0 s (29.6% faster), others -0.05 s	PASS
ST-05	M3	closure - static routing strands vehicles, re-routing delivers 100%; slowdown - broadcast >= 20% faster than local	closure: static 0%, broadcast 100%; slowdown: local 144.61 s vs broadcast 95.47 s	PASS
ST-06	M4	TSP makes buses faster, rejects some proposals under heavy cross traffic, and saves passenger time overall	heavy cross: 21 accepted / 11 rejected, net 525 passenger-s saved	PASS
ST-07	M3	A* is 100% optimal and expands fewer nodes than UCS; Greedy is not always optimal	A* 14.5 vs UCS 35.5 nodes; greedy optimal 54.0%	PASS
ST-08	M5	hill-climbing-tuned parameters beat the defaults on unseen validation traffic	validation wait 8.65 s -> 6.96 s; params pushed to agents	PASS
ST-09	M1	2x2-4x4 grids all clear, Max-Pressure beats Fixed-Time at every size	2x2: 45.4% less wait, 0.12 ms/tick; 3x3: 44.5% less wait, 0.24 ms/tick; 4x4: 42.4% less wait, 0.38 ms/tick	PASS

4. Scenario Test Set-up

ID	Network	Demand	Seeds	Baseline vs treatment	Metric
S1	2x2 grid	Poisson 0.35 veh/s for 300 s, 12 O-D pairs	10	Fixed-Time 20 s vs Max-Pressure (+/- green-wave)	avg / p95 wait, travel, stops, fuel
DM	3x3 demo, 6 tracked cars	accident closes J01-J11 at t = 10 s	1	original vs new route	re-routed cars, arrival
S2	2x2 grid	0.45 veh/s N-S, 0.07 veh/s E-W	5	Fixed vs MP without vs with anti-starvation	wait by axis, starved cars
S3	2x2 grid	0.3 veh/s background + 1 ambulance S0->N0	10	same MP controller, preemption off vs on	ambulance travel & wait; others' wait
S4	2x2 grid	12 W0->E0 cars + 0.15 veh/s background; incident at t = 10 s	5	static vs local vs broadcast; closure and slowdown	arrived %, travel time
S5	2x2 grid	4 buses W0->E0 + 0.2 or 0.5 veh/s N-S cars	8	TSP off vs on	bus travel, car wait, accept/reject, passenger-s
S6	2x2 grid	asymmetric 0.4 veh/s	3 train + 5 validation	default vs hill-climbed parameters	avg wait on unseen seeds
S7	6x6 grid, random loads, 1 closed road	200 random trips	1	BFS/UCS/Greedy/A*/W-A*	expansions, optimality
S8	2x2 to 8x8	0.18 x n veh/s, boundary O-D	1	Fixed vs MP at each size	wait, ms per tick

5. Defect Log

Defects found while reviewing and testing the original code base, and during this test cycle. Each fix is protected by a regression test.

ID	Defect	Impact	Fix	Regression test
DEF-01	IntersectionAgent.reason() ran twice per tick (called by step() and again by act())	Phase timers ran at 2x speed: 5 s min-green was really 2.5 s; emergency hold halved	BaseAgent.step() stores the decision once; act() applies it	UT-17, IT-02
DEF-02	Vehicle departure_time was ignored	All vehicles entered at t = 1; 'staggered' demand in every scenario was meaningless	Vehicles stay off the road until their departure time	UT-22
DEF-03	Ambulance baseline used Fixed-Time while the treatment used Max-Pressure	Reported 44% gain mixed two effects	Both runs use the same controller; only preemption changes (now 29.6%)	ST-04
DEF-04	Preemption jumped straight from one green to the other	Unsafe: no yellow clearance	All switches go through a 3 s yellow, including emergencies	IT-02
DEF-05	Ambulance re-sent PREEMPT_OVERRIDE every tick, including to non-signalised nodes	Message flood; junctions never released	One request per junction inside a 15 s ETA window; CANCEL after crossing	IT-01
DEF-06	Green-wave INFORM received but ignored (`pass`)	Claimed cooperation did nothing	Neighbour extends the matching green for 25 s	IT-07
DEF-07	Re-planning while on a road started from that road's start node	New path could disagree with the road being driven	Plan from the road's end node and prepend the current road	IT-05
DEF-08	Scenario 'tests' had no assertions; unfinished vehicles were dropped from averages	Tests could not fail; results were biased	Run until clear; acceptance asserts	ST-01..09
DEF-09	Vehicles waiting to depart ignored INCIDENT_ALERT until they started driving (found by IT-05 failing)	Some vehicles still planned through a closed road	Pre-trip listening: messages processed before departure	IT-05
DEF-10	Vehicles routed on the true road state (omniscient)	TMC broadcast carried no information; S4 local = broadcast	Per-vehicle beliefs (known_blocked, known_speed)	UT-23, ST-05

ID	Defect	Impact	Fix	Regression test
DEF-11	Coordinator sent CONGESTION_WARNING every tick; index diluted by empty roads	Broadcast storm; warnings rarely triggered	20 s rate limit; index over occupied roads	IT-09 (message counts stable)

5.1 Negative results kept on record

- Green-wave hand-off on random grid demand: avg wait 8.11 s without vs 9.26 s with. Not a defect: platoon hand-off needs predictable corridor flow, so it is optional and its message exchange is verified by IT-07.
- Bus priority roughly doubles cross-street car wait. Accepted as a design trade-off, because net passenger-seconds improve (ST-06).
- With a full road closure, local sensing and TMC broadcast give identical results on a grid (equal-length detours). The broadcast's value is shown with the slowdown case (ST-05).

6. Conclusion

> **43 of 43 tests pass.** Every functional claim in the report is covered by at least one test with a measurable criterion. The safety invariant (no green-to-green switch, yellow of at least 3 s) holds across ambulance and bus interactions. Eleven defects in the original code base were found and fixed, and each fix has a regression test.